1. Container
2. Plates
3. Separator
4. Cell cover
5. Electrolyte
6. Terminal post

Container

The container is a single piece moulded part made up of hard rubber or bituminous. The container will house all the internal parts of a battery and holds the liquid electrolyte. The container is divided into six compartments for the six cells. Projections are provided on the inside at the bottom to support the plates. Bridges are used to prevent the plate from touching at the bottom.

Plates

Plates are of two types viz. 1. Positive plate and 2. Negative plate

The positive plate is made of lead peroxide. This is dark brown, hard and brittle substance. Lead oxide acts as active material. Highly porous in nature. The negative plate is made of pure lead in soft sponge condition. Spongy lead acts as active material. Grey in colour. Highly porous in nature. Grids are used to hold the plates and to conduct the electrical charge.

Separator

Separators are placed between positive plates and negative plates to insulate them between each other. This will prevent the short-circuiting between the positive and negative plates. The separators are thin sheets of non-conducting material made up of chemically treated wood, porous rubbers, or mats of glass fibre. The separators must be porous so that the electrolyte may circulate between the plates.

Cell cover

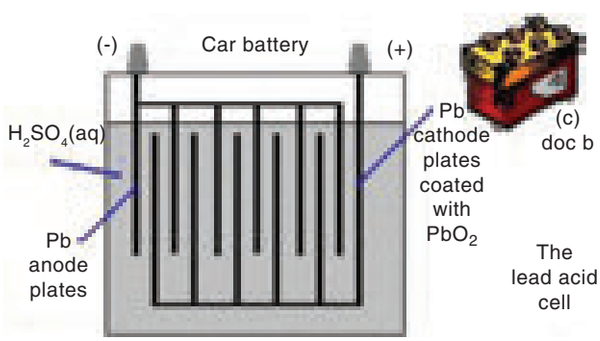
Cellcover is used to cover the complete cell. It protects the cell from the dust as well as other external impurities. Vent holes are provided to exhaust the gases generated in the cell to the atmosphere. Filler openings are available to fill up the electrolyte.

Electrolyte

In lead-acid battery dilute sulphuric acid (H_2SO_4)is used as an electrolyte. For this purpose, one part concentrated sulphuric acid is mixed with three parts of distilled water.

Terminals

Terminals are used to connect the charging circuit as well as the load. For easy identification, the diameter of the positive terminal is bigger than the negative terminal.



Working principle

During discharging (i.e., supplying a current), atoms from the spongy lead on

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